# Praveen Samarasekara

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## Summary

A passionate undergraduate with a strong interest in artificial intelligence, signal processing, and embedded systems. Always eager to explore new technologies and solve real-world problems through intelligent systems. I am actively seeking opportunities to apply my skills in research, industry, or project-based environments.

## Education

**Bachelor of Science (Hons) in Engineering** — *University of Moratuwa* *Jan 2024 – Present*
*Specializing in Electronic & Telecommunication Engineering*

- CGPA: 3.59 / 4.0 *(Dean's List — Semester 1, Semester 4)*

**G.C.E. Advanced Level** — *Bandaranayake Central College Veyangoda* *2020*
*Physical Science Stream*

- 3A passes Z-Score: 2.5003 District Rank: 23

## Projects

**Customer Churn Prediction using Machine Learning(Ongoing)** GitHub

- **Tools:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook
- Developed a machine learning model to predict customer churn using a publicly available Kaggle dataset and performed comprehensive data preprocessing, feature engineering, and exploratory data analysis.
- Trained and evaluated a Random Forest classifier, achieving strong predictive performance through hyperparameter tuning and cross-validation.
- Analyzed feature importance to identify the key factors influencing customer retention and presented insights using data visualizations.

**PrintEase Solutions – Automated Self-Service Printing Kiosk *(Frontend and Circuit design)*** GitHub

- **Tools:** React, Django, PostgreSQL, ESP32, PayHere API, Vite
- Developed an automated self-service printing kiosk that enables users to upload documents, complete secure online payments, and print without operator assistance.
- Integrated a React frontend, Django backend, and ESP32-based hardware interface to automate the document printing workflow and improve customer convenience.

**Music Genre Classification Engine using PySpark MLlib *(Ongoing)***

- **Tools:** Python, Apache Spark, PySpark MLlib, Flask, Docker, HTML, CSS, JavaScript
- Engineered a scalable end-to-end Big Data machine learning pipeline using PySpark to classify music genres from song lyrics, covering data preprocessing, feature engineering, model training, and evaluation.
- Implemented a multi-model voting ensemble combining Gradient-Boosted Trees (One-vs-Rest), Naive Bayes, and Logistic Regression to improve classification performance.
- Developed a Flask web application with a REST API to serve the trained ensemble model, enabling real-time genre prediction through an interactive web interface.
- Containerized the complete application, including the Apache Spark environment and all dependencies, using Docker for reproducible, one-command deployment.

**University Lost and Found System** GitHub

- **Tools:** Java, Node.js, npm
- Built a full-stack platform for reporting and reclaiming lost items on campus, with a Java backend and a Node.js/npm-based frontend.
- Worked on the REST API layer as well as the frontend, connecting the client to backend services for reporting, searching, and claiming lost items.

**Multichannel Temperature Data Logger for Vaccine Cold Chain Transport**

- **Tools:** Embedded C, RTC, SD Card Module, Wireless Communication Module, Microcontroller, Alert Interfaces
- Designed and developed a multichannel temperature data logger for monitoring vaccine cold chain transport, using a Human-Centered Design (HCD) approach.
- Implemented real-time clock (RTC) timestamping, SD card data storage, and an alert interface to flag temperature excursions across multiple channels.
- Added wireless connectivity to transmit temperature data and alerts remotely, enabling real-time monitoring during transport.

**Tri-Mode Class AB Power Amplifier**

- **Tools:** Analog Circuit Design, LTspice, Altium Designer, Power Electronics
- Designed and built an industrial-grade Class AB audio power amplifier using discrete analog components with support for Mono, Stereo, and Bridge (BTL) operating modes.
- Developed the complete analog signal chain, including the power stage, biasing network, thermal stability, and protection circuitry to achieve high-fidelity audio performance.

## Technical Skills

**Languages:** C, C++, Python, MATLAB, Java, JavaScript

**Embedded / Hardware:** STM32, ESP32, Arduino, Raspberry Pi, Altium Designer

**Frameworks & Tools:** TensorFlow, PyTorch, OpenCV, NumPy, Git, GitHub, Docker

**Software:** MATLAB, Simulink, Altium, SolidWorks, VS Code

**Operating Systems:** Linux, Windows

**Markup:** LaTeX, Markdown, HTML, CSS

## Honors & Awards

- Dean's List Placement — Semesters 1 — University of Moratuwa *2024, 2025*
- Merit Pass — G.C.E. Advanced Level Examination *2020*

## Competitions & Achievements

- **Participant** — Sri Lanka Robotics Challenge (SLRC) 2025: Contributed to the design, development, and testing of an autonomous mobile robot for robotics competition tasks.

## Volunteering

**Committee Member** — *Classical Music Society, University of Moratuwa* *2025 – Present*

- Serve as a committee member, contributing to the planning and organization of concerts, workshops, and society events.
- Perform as a violinist in society concerts and musical events, promoting classical music within the university community.

**Editor** — *E-Carrier Annual Magazine, Department of Electronic and Telecommunication Engineering, University of Moratuwa* *2025 – Present*

- Served as an editor for the department's annual magazine, reviewing, editing, and organizing technical and non-technical content for publication.

## Courses & Certifications

- **Supervised Machine Learning: Regression and Classification** — *DeepLearning.AI & Stanford Online (Coursera)* 2026